

A-MES

AUTOMOTIVE MES • SEOYON E-HWA

VOL 01

Technology Stack & Architecture

.NET 9 • Blazor Server • WinForms • MAUI • MSSQL

Document • L3 Detailed Design Specification

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TABLE OF CONTENTS

Technology Stack & Architecture

Tech_Stack A-MES · VOL.1 · Technology Stack

TECHNOLOGY STACK

Implementation stack for the A-MES three-system architecture — **Office Web**, **Shop-Floor POP**, and **Logistics PDA**. All components target the **.NET 8/9** runtime and **Microsoft SQL Server**. Development is standardized on **Visual Studio 2022** with optional Rider/VS Code support. The implementation team is composed of .NET / MSSQL specialists; tooling choices favor long-term familiarity, native debugging, and minimal external-runtime dependencies.

OFFICE WEB

Blazor Server

.NET 8/9 + SignalR

SHOP-FLOOR POP

WinForms

.NET 8 (Designer)

LOGISTICS PDA

.NET MAUI

Android Native

DATA

SP + ADO.NET

MSSQL 2019+

AUTH

ASP.NET Identity

+ JWT for API

▼ ARCHITECTURE OVERVIEW

⊕ THREE-SYSTEM ARCHITECTURE · SHARED API · MSSQL



OFFICE WEB

Planner / Engineer · Desktop browser

Blazor Server

.NET 9

SignalR

MudBlazor



SHOP-FLOOR POP

Operator · Fixed touch terminal

WinForms

.NET 8

HttpClient

ClickOnce



LOGISTICS PDA

Warehouse · Android handheld

.NET MAUI

Android 11+

Zebra DW

SQLite cache

↓ ALL THREE SHARE THIS BACKEND ↓

SHARED API · ASP.NET CORE WEB API

Single Web API project — all three clients consume it

ASP.NET Core 9

Minimal API / Controllers

JWT bearer

SignalR hub

Serilog

DATABASE

MSSQL Server

MSSQL 2019/2022

Stored Procs

IDENTITY

User store + role-based authorization

ASP.NET Identity

ADO.NET

AspNetUsers

Roles

Why three separate clients? Office workers need a rich data-grid UI on large monitors (Web). Operators need large touch targets on fixed terminals (POP) without browser chrome. Logistics workers need a handheld with barcode scanner integration and offline tolerance (PDA). A single UI cannot serve all three well, so we share the *back-end* (API + DB + Identity) and tailor each *front-end* to its environment.

▼ SYSTEM 1 · OFFICE WEB



SYS.1 · OFFICE WEB · 74 SCREENS

Blazor Server on .NET 9

PP · MNT · RPT · SYS · MD modules · Desktop browser (1920px)

LAYER	SELECTION	RATIONALE
Runtime	.NET 9 (LTS-class)	Latest stable runtime, free upgrade path from .NET 8 LTS. Compatible with WinForms POP and MAUI PDA — single SDK across stack.
Framework	Blazor Server	C# end-to-end (no JavaScript framework). Component model resembles WinForms — knowledge transfer to POP team. SignalR built in for live dashboards (PP-OEE, PP-ODM, RPT-07, MNT-09). Hot Reload + IntelliSense + native VS debugger.
UI Library	MudBlazor 6.x (free, MIT)	Material-design components. Includes MudDataGrid (sort/filter/paging out of the box for 14+ table-heavy screens), MudChart (Pareto/OEE), MudDatePicker, theming. Alternative: Syncfusion Blazor (commercial, richer Gantt/scheduler if PP-LSB or MNT-05 PM calendar needs it).
Real-time	SignalR (Blazor Server intrinsic)	Every Blazor Server circuit already uses SignalR — pushing live updates to live dashboards (OEE/Andon/alarms) requires zero extra plumbing.
Auth UI	ASP.NET Identity + Razor pages	Login/logout/password reset via scaffolded Identity UI. Custom branding overlay matching A-MES.
Charts	MudBlazor Chart, ChartJs.Blazor (fallback)	MudBlazor's built-in chart handles 80% of cases (line/bar/pie/donut). For complex Pareto or stacked OEE, ChartJs.Blazor wraps the de-facto JS chart library while keeping the binding in C#.
Reporting (PDF)	QuestPDF (free for < \$1M ARR) or Telerik Reporting	Fluent C# API for generating delivery notes, BOL, daily production reports. Embeds smoothly in Blazor — no JS required.
Excel I/O	ClosedXML	For RPT export / SYS-05 audit export / MD bulk import. Free, MIT.
Background jobs	Hangfire	For PP-02 SAP nightly imports, RPT scheduled emails (SYS-06 Notification), daily snapshots. Dashboard UI included.
Logging	Serilog + Seq (optional)	Structured logging. Audit Log (SYS-05) writes to MSSQL via dedicated SP. Application logs to console + rolling file.
Hosting	Kestrel behind IIS on Windows Server 2022	IIS is the standard for on-prem .NET deployments in Korean enterprise. Windows Authentication available if cross-checking with AD becomes a future requirement.

Blazor Server tradeoff to plan for: each connected user holds an open WebSocket (SignalR circuit) to the server. For ~200 concurrent office users this is well within a single server's capacity (~100MB RAM ceiling for sessions). If concurrent users

approach 1,000+, add a load balancer with sticky sessions or move to *Blazor WebAssembly + Web API* later.

▼ SYSTEM 2 · SHOP-FLOOR POP



SYS.2 · SHOP-FLOOR POP · 33 SCREENS

WinForms on .NET 8

INJ · IMG · PNT · QC processes · Fixed touch terminals on the line

LAYER	SELECTION	RATIONALE
Runtime	.NET 8 (LTS until Nov 2026)	LTS recommended for long-lived fixed terminals. Designer support in Visual Studio is best in .NET 8.
Framework	WinForms (.NET 8)	Modern WinForms preserves the classic designer experience but runs on cross-cutting modern .NET. Drag-and-drop UI design matches what the team already knows.
Touch UI	Custom large-button styling + HighDpiMode.PerMonitorV2	Buttons ≥ 64px tall, font ≥ 16pt, dark theme (matches POP demo). Avoid system dialogs — use full-screen kiosk shell.
API client	HttpClient + System.Text.Json	All data goes through the shared Web API — POP never talks to MSSQL directly. Polly retry policy on transient network errors.
Auth	JWT bearer (operator employee ID + 4-digit PIN, or RFID badge)	Short-lived JWT (8 hr shift). RFID badge tap triggers /auth/badge endpoint. PIN keypad on screen for password.
Barcode	HID keyboard-wedge scanners (default)	Scanned barcode appears as typed text in focused input. No SDK needed. For 2D imager features, Honeywell/Zebra SDKs available as optional add-on.
Charts (live)	LiveCharts2 (free, MIT)	For INJ-07 Production Status mini-charts on the POP. Smooth WinForms integration via CartesianChart control.
Andon (INJ-08)	SignalR client (Microsoft.AspNetCore.SignalR.Client)	POP connects to the same SignalR hub the Office Web uses. Andon call is pushed to all subscribed dashboards within ≤ 200 ms.
Offline cache (light)	Local JSON file (per workstation)	Master data (work-order list, mold list, defect codes) cached locally so a brief network blip doesn't block production entry. Posts queue and retry on reconnect.
Logging	Serilog + rolling file (local) + Sink.Http to API	Local file survives WAN outage; HTTP sink ships logs centrally when online.
Deployment	ClickOnce from internal IIS share	Each POP terminal auto-updates from a single network publish URL. No per-terminal manual install. Roll back by re-publishing previous version.

▼ SYSTEM 3 • LOGISTICS PDA

SYS.3 • LOGISTICS PDA • 18 SCREENS

.NET MAUI (Android)



LAYER	SELECTION	RATIONALE
Runtime	.NET 8 (MAUI)	LTS. Single C# codebase compiles to native Android APK (and iOS/Windows if needed later).
Target	Android 11 (API 30) minimum, target Android 13 (API 33)	Covers Zebra TC2x/TC5x, Honeywell CT45, Datalogic Memor 11/12, etc. Most Android-based industrial handhelds shipped 2022+ support API 30.
Framework	.NET MAUI (XAML or Blazor Hybrid)	Recommend XAML pages for the 420px mobile-frame layout demoed. Blazor Hybrid is an option if reusing Office Web's Razor components, but XAML gives faster native start-up on lower-end handhelds.
Barcode scan	Zebra DataWedge intent / Honeywell IntentApi	Hardware scan button on the handheld emits an Android intent (configurable via vendor SDK). MAUI listens for the intent and routes to the active page's scan handler. No 3rd-party library required.
RFID (FG-01 etc.)	Vendor RFID SDK (Zebra RFID SDK / Bluebird etc.) wrapped in MAUI binding library	Some FG/WH workflows need UHF RFID tag reads. Vendor SDK is invoked via Android binding generated by MAUI tooling.
Offline cache	SQLite via sqlite-net-pcl	PDA must continue working in the warehouse where Wi-Fi can drop. Inbound/picking transactions queue locally and sync on reconnect. SQLite is the standard Android local store.
API client	HttpClient + Polly retry/jitter	All data exchanged with the shared Web API. Retry policy with backoff for spotty wi-fi.
Auth	JWT bearer (employee ID + PIN or RFID badge)	Same flow as POP. Token stored in secure-storage SecureStorage.SetAsync.
Print (FG-06 B0L)	Bluetooth or network ZPL to Zebra ZD421 / ZQ520	ZPL command stream over TCP socket. For mobile printers, use Bluetooth via Plugin.BLE.
Push notifications	Firebase Cloud Messaging (optional)	For pushing "urgent picking" or "QC hold" alerts to the PDA out-of-hours. Skip if not required.
Deployment	APK side-loaded via MDM (SOTI MobiControl / VMware Workspace ONE) or Zebra StageNow	Internal app — not published to Play Store. MDM pushes new APK to every fielded handheld on tagged updates.



SHARED • BACKEND API

ASP.NET Core Web API

One project, three clients — single source of business logic

LAYER	SELECTION	RATIONALE
Runtime / FX	ASP.NET Core 9 (Web API)	Hosted under Kestrel + IIS reverse proxy. Same Visual Studio solution as Office Web project for development convenience.
API style	Controller-based REST + selective Minimal API endpoints	Controllers for resource CRUD (familiar pattern). Minimal API for high-frequency utility endpoints (e.g., scan validation).
Data access	Stored Procedures + ADO.NET (SqlConnection / SqlCommand)	Per team decision: business rules live in stored procedures (DBA-managed, transactionally explicit). C# layer is thin SP invoker. Use Dapper as a lightweight mapper if the SP call/result mapping becomes repetitive (still SP-driven — Dapper just maps the result rows).
DB schema mgmt	SSDT (SQL Server Database Project) + DACPAC	Schema is checked into the Git repo as a .sqlproj. Deploy compares the DACPAC to the target DB and generates upgrade script. Code-review friendly. No EF migrations needed (we're SP-driven).
Identity	ASP.NET Identity	User accounts stored in AspNetUsers/AspNetRoles tables in the same MSSQL. Password hashed (PBKDF2). Role-based authorization via [Authorize(Roles=...)] attributes. SYS-01 (User Mgmt) and SYS-02 (RBAC) screens manipulate these tables.
Token issuance	JWT bearer for POP / PDA / external API clients	Web API exposes /auth/login that returns a JWT for non-browser clients. Office Web uses Identity cookies natively. Token includes role claims and a short expiry (8 hr).
Validation	FluentValidation	Declarative request validation, decoupled from the controller. Easier code review than DataAnnotations attributes scattered in DTOs.
Real-time	SignalR Hub (1 hub, multiple groups)	Single hub /hubs/realtime. Groups: oee.line-{id}, andon.plant-{id}, alarm.{user}. Office Web, POP, PDA all subscribe.
Background jobs	Hangfire (MSSQL storage)	SAP nightly import (PP-02), report subscriptions (SYS-06), PM-due notifications (MNT-05), daily snapshots. Web dashboard for monitoring jobs.
Logging	Serilog → file + MSSQL + Seq (optional)	Structured logs (JSON). SYS-05 Audit Log writes to a separate SP_AUDIT_INSERT for compliance — kept distinct from app logs.
OpenAPI	Swashbuckle (Swagger UI at /swagger)	Auto-generated API reference for developers. Disable in production or restrict by role.

▼ AUTHENTICATION & AUTHORIZATION FLOW

- 1 Web user login** · Office Web user types ID/password in scaffolded Identity Razor Page. Identity validates against `AspNetUsers` (PBKDF2 hash). On success, sets `.AspNetCore.Identity.Application` cookie. Cookie carries role claims via `ClaimsPrincipal`.
- 2 POP / PDA login** · Operator scans employee ID and enters 4-digit PIN (or taps RFID badge). Client POSTs to `/api/auth/login` with credentials → API verifies via Identity's `SignInManager.CheckPasswordSignInAsync` → returns a **JWT** with `sub`, `role`, `shift`, `line` claims and 8-hour `exp`.
- 3 Per-request authorization** · Each API endpoint declares `[Authorize(Roles = "Operator,Supervisor")]` or `[Authorize(Policy = "QC.Approve")]`. Custom policies map to RBAC permission matrix maintained in SYS-02. Identity middleware enforces before controller runs.
- 4 SignalR auth** · WebSocket upgrade carries the bearer token via query string (`?access_token=...`) for POP/PDA, or the Identity cookie for Office Web. Hub `OnConnectedAsync` reads claims and adds the connection to the correct group (e.g., `oee.line-INJ-01`).
- 5 Audit logging** · Every authenticated mutation invokes `IAuditLogger.LogAsync(userId, action, target, ipAddress)` middleware → writes to `tbl_AuditLog` via `SP_AUDIT_INSERT`. Visible in SYS-05.
- 6 Refresh / revoke** · JWT does not auto-refresh — operators re-sign-in at shift change. To revoke before expiry (e.g., user disabled in SYS-01), token is added to a small in-memory blocklist checked by a custom `ITokenValidator` on each request.

▼ INTEGRATION POINTS · EXTERNAL SYSTEMS

EXTERNAL SYSTEM	A-MES TOUCH-POINT	LIBRARY / PROTOCOL	NOTES
SAP B1	PP-02 Supply Plan Import · MD-07 Vendor sync · FG outbound posting	SAP DI-API (COM) or Service Layer REST	REST preferred for cross-platform. Hangfire job runs nightly at 02:00 EST pulling Sales Orders → A-MES forecasts.
RFID Reader (FG/WH)	FG-01 Stocking · WH-02 Inbound	Vendor SDK (Zebra/Impinj) or TCP/IP raw stream · MQTT	PDA reads via on-device UHF module. Fixed-station readers stream tags via MQTT to API.
OPC-UA (equipment)	MNT-03 OEE · PP-OEE · INJ shot count	OPCFoundation.NetStandard.Opc.Ua (free, open-source)	Background subscription service polls/subscribes to PLC tags (Siemens, Mitsubishi, AB). Writes to tbl_EquipmentSignal.
Barcode scanners	All POP/PDA workflows	HID keyboard wedge (default) · Zebra DataWedge intent	No SDK for HID. For 2D/PDF417/DPM features use vendor SDK. PDA always uses DataWedge intent.
Label printers	PNT-07 Label Apply · FG-06 BOL	Zebra ZPL via TCP 9100 · BPL for stock-label · PDF for BOL	Send raw ZPL string over TCP socket. For Honeywell, Direct Printer Language. Mobile printer (FG-06) over Bluetooth.
Email (SYS-06 / RPT subscriptions)	Alerts · scheduled reports	MailKit over corporate SMTP	MailKit is the modern replacement for Smtplib. Works with O365 / Gmail Workspace / on-prem Exchange.
SMS	SYS-06 critical alerts	Twilio .NET SDK (or domestic gateway)	U.S. context — Twilio is the default. For Korean deployment, swap to Aligo/SENS.
Microsoft Teams	SYS-06 channel notifications	Incoming Webhook (HTTPS POST JSON)	No SDK needed — simple webhook URL per channel.
Active Directory (optional)	Future SSO / user provisioning	Microsoft.Identity.Web + LDAP fallback	Not in v1. ASP.NET Identity is self-contained for launch; AD federation can be added without rewriting auth code.

▼ DEVELOPMENT ENVIRONMENT

Toolchain — Visual Studio 2022 standard

Every team member uses the same Visual Studio 2022 installation profile to ensure consistent .editorconfig, debugger settings, and template availability.

✓ **Visual Studio 2022 Community/Pro** v17.10+ — install workloads: `ASP.NET and web development`, `.NET`

✓ **SQL Server Management Studio (SSMS) 20** — for SP development and DBA tasks

desktop development , .NET Multi-platform App UI development , Data storage and processing

✓ **SQL Server 2022 Developer Edition** (local) — free, full feature parity with production for local dev

✓ **Android SDK + emulator** (via VS MAUI workload) — for PDA development without a physical handheld

✓ **Hot Reload** enabled for Blazor Server and WinForms — instant feedback during UI iteration

✓ **Git for Windows** + GitHub or Azure DevOps — repo: `github.com/MES-TAEHEE/ames-docs`

✓ **Postman or Bruno** — REST API testing against the shared Web API

✓ **.editorconfig** in repo root — enforced C# code style (4-space indent, brace placement, file-scoped namespaces)

▼ DEPLOYMENT TOPOLOGY



OFFICE WEB

Host: Windows Server 2022 + IIS
10 reverse-proxy → Kestrel

SSL: Wildcard cert on IIS

URL:

`https://ames.seyonehwa.us`

Scale: 1 server initially; sticky-session LB if > 500 concurrent

Update: Web Deploy package via Azure DevOps release pipeline



SHOP-FLOOR POP

Host: Each line has a fixed Windows 11 IoT terminal (touch monitor)

Distribution: **ClickOnce** published to internal IIS share

URL:

`https://ames.seyonehwa.us/pop/install`

Update: Terminal checks for update on launch; auto-installs new version

Rollback: Re-publish previous version



LOGISTICS PDA

Device: Zebra TC52ax /
Honeywell CT45 (Android 13)

Distribution: Signed APK pushed via MDM (SOTI / Workspace ONE / Zebra StageNow)

Update: MDM tag drives version rollout (canary → 10% → 100%)

Offline: Local SQLite cache; transactions queue and sync on Wi-Fi recovery

Crash reporting: AppCenter Diagnostics (optional)

CI/CD pipeline: All three projects build in a single Azure DevOps pipeline (or GitHub Actions). Triggers on push to main: **(1)** Web API + Web App → Web Deploy package → IIS, **(2)** WinForms POP → ClickOnce publish, **(3)** MAUI Android → signed APK uploaded to MDM endpoint. DACPAC for the database is applied to staging on PR merge, to production on tagged release.

▼ COMPLETE LIBRARY MATRIX

UI • OFFICE WEB

MudBlazor

UI • OFFICE WEB (COMMERCIAL ALT.)

Syncfusion Blazor

Material-design Blazor component library. Includes DataGrid (sort/filter/paging), Chart, DatePicker, Dialog, Theming.

FREE · MIT v7.x

Richer enterprise components (Gantt, Scheduler, RichTextEditor). Adopt if MudBlazor falls short for PP-LSB Gantt or MNT-05 PM calendar.

COMMERCIAL OPTIONAL

DATA · SP RUNNER / MAPPER

Dapper

Lightweight result-set mapper. Use to map SP result rows to C# DTOs (still SP-driven).

FREE · Apache

DATA · SCHEMA MGMT

SQL Server Database Project (SSDT) / DACPAC

Schema-as-code in the repo. Generates upgrade scripts on deploy.

FREE · Microsoft

API · VALIDATION

FluentValidation

Declarative request validation with C# expressions.

FREE · Apache

API · OPENAPI

Swashbuckle.AspNetCore

Swagger UI at /swagger for testing.

FREE · MIT

BACKGROUND JOBS

Hangfire

Recurring/delayed/queue jobs with MSSQL persistence. Web dashboard included.

FREE · LGPL

LOGGING

Serilog + Sinks (File / MSSQL / Seq)

Structured logging with multiple sinks.

FREE · Apache

PDA · DB

sqlite-net-pcl

Offline cache on the handheld. Sync queue when Wi-Fi recovers.

FREE · MIT

PDA · HTTP RESILIENCE

Polly

Retry, circuit-breaker, jitter for unreliable Wi-Fi.

FREE · BSD

POP · CHARTS

LiveCharts2

Real-time WinForms charts for cycle time, OEE on the line.

FREE · MIT

PDF

QuestPDF

Fluent C# API for BOL, delivery notes, daily reports.

FREE < \$1M ARR

EXCEL

ClosedXML

Excel export/import (RPT-09 Report Center · MD bulk import).

FREE · MIT

EMAIL

MailKit

Modern SMTP/IMAP client. Replaces deprecated Smtplib.

FREE · MIT

OPC-UA (EQUIPMENT)

SAP B1 (OPTIONAL)

OPCFoundation.NetStandard.Opc.Ua

Official .NET OPC-UA client. Subscribes to PLC tags.

FREE · MIT

SAP B1 Service Layer SDK (REST)

REST + OData against SAP B1. Plain HttpClient works.

SAP-LICENSED

▼ SOLUTION STRUCTURE

Recommended single Visual Studio solution layout — one .sln spans all three clients + shared API + DB project, enabling F5-to-debug across the stack.

```
A-MES.sln
├── 01_Shared/
│   ├── AMES.Contracts/           # DTOs, enums, shared between API and clients
│   └── AMES.Common/              # JWT helpers, validation, audit interfaces
│
├── 02_Database/
│   └── AMES.Database/            # SSDT .sqlproj – schema, SPs, seed data → DACPAC
│
├── 03_Api/
│   └── AMES.Api/                 # ASP.NET Core Web API + SignalR hub + Hangfire
│
├── 04_OfficeWeb/
│   └── AMES.OfficeWeb/          # Blazor Server (74 screens) + ASP.NET Identity UI
│
├── 05_PopWinForms/
│   └── AMES.Pop/                # WinForms (.NET 8) for POP terminals
│
├── 06_PdaMaui/
│   └── AMES.Pda/                # .NET MAUI (Android) for handheld PDA
│
└── 07_Tests/
    ├── AMES.Api.Tests/          # xUnit + WebApplicationFactory
    ├── AMES.OfficeWeb.Tests/    # bUnit for Blazor components
    └── AMES.Integration.Tests/  # End-to-end (testcontainers MSSQL)
```

Why one solution? Allows F5 debugging across boundaries (e.g., set breakpoint in API, drive from Blazor Web). Single dotnet build validates all clients still compile against the shared contracts. Each project still publishes independently.